

---

# Expectation and Midterm 1

Antonio de Leon

Probability and Statistics for Engineers   Fall 2026

|                       |                                          |
|-----------------------|------------------------------------------|
| Monday, October 12    | Expectation, linearity, and indicators   |
| Wednesday, October 14 | Review through the binomial distribution |
| Friday, October 16    | Midterm 1                                |

# Expectation and the examination boundary

## Today

Compute a mean, derive linearity, and use indicators to obtain a count's mean.

## Midterm 1

Required material through October 9.

Expectation, variance, and joint distributions are excluded.

**Recall the model.** Three independent fair coin tosses;  $X$  counts heads.

| $x$      | 0   | 1   | 2   | 3   |
|----------|-----|-----|-----|-----|
| $p_X(x)$ | 1/8 | 3/8 | 3/8 | 1/8 |

Why does  $X = 1$  have three times the probability of  $X = 0$ ?

## Expected value

### Definition for a finite-valued random variable

$$\mu = \mathbb{E}[X] = \sum_{x \in S_X} x p_X(x), \quad S_X = \{x : p_X(x) > 0\}.$$

Each possible value receives its probability as a weight.

$$p_X(x) \geq 0, \quad \sum_{x \in S_X} p_X(x) = 1.$$

**A useful check.** If  $a \leq X \leq b$ , why must  $a \leq \mathbb{E}[X] \leq b$ ?

---

$X$  is a random variable;  $\mathbb{E}[X]$  is a number determined by its distribution.

For countably many values, we assume  $\sum_x |x| p_X(x) < \infty$ . All required examples today have finitely many values.

# Mean of a three-toss count

## Values and their weights

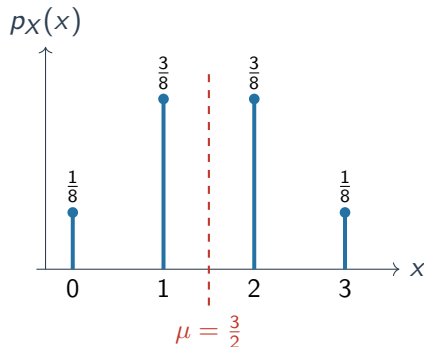
|          |       |       |       |       |
|----------|-------|-------|-------|-------|
| $x$      | 0     | 1     | 2     | 3     |
| $p_X(x)$ | $1/8$ | $3/8$ | $3/8$ | $1/8$ |

Write each contribution  $x p_X(x)$ , then add.

---

---

---



The mean is 1.5 heads, although the count is never 1.5.

An expectation summarizes a probability model. An observed average is calculated from data.

## Grouping outcomes by value

On a finite sample space, let  $q_\omega = P(\{\omega\})$ . These weights need not be equal.

### One mass combines outcome weights

| Outcomes      | $X(\omega)$ |
|---------------|-------------|
| TTT           | 0           |
| HTT, THT, TTH | 1           |
| HHT, HTH, THH | 2           |
| HHH           | 3           |

$$p_X(x) = \sum_{\omega: X(\omega)=x} q_\omega.$$

The three one-head outcomes each contribute  $1 \cdot (1/8)$  to the mean.

Fair tosses:  $q_\omega = 1/8$ .

Both sums collect the same terms, grouped in different ways.

## Linearity of expectation

For integrable  $X, Y$  and real constants  $a, b$

$$\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y].$$

**We prove the finite-sample-space case.** Both variables use the same outcome  $\omega$ .

$$Z(\omega) = aX(\omega) + bY(\omega).$$

By the outcome representation,

$$\mathbb{E}[Z] = \sum_{\omega \in \Omega} Z(\omega)q_{\omega}.$$

**Substitute the definition of  $Z$  and distribute the weights.**

---

Integrable means  $\mathbb{E}[|X|], \mathbb{E}[|Y|] < \infty$ . This holds automatically for the finite models here.

## Linearity on a finite sample space

We have

$$\mathbb{E}[aX + bY] = \sum_{\omega \in \Omega} [aX(\omega) + bY(\omega)] q_{\omega}.$$

Separate the two sums and identify each expectation.

---

---

---

Repeated application gives the finite-sum version

$$\mathbb{E}\left[\sum_{i=1}^n a_i X_i\right] = \sum_{i=1}^n a_i \mathbb{E}[X_i].$$

## Constants and affine changes

A constant random variable has mean

$$\mathbb{E}[c] = \sum_{\omega \in \Omega} c q_{\omega} = c \sum_{\omega \in \Omega} q_{\omega} = c.$$

For integrable  $X$  and real constants  $a, b$

$$\mathbb{E}[aX + b] = a\mathbb{E}[X] + b.$$

**The same coin experiment.** Score  $+1$  for a head and  $-1$  for a tail. If  $X$  counts heads in three tosses, the total score is  $Y = 2X - 3$ .

## Mean of an indicator

For an event  $A$ , its indicator records whether  $A$  occurs:

$$I_A(\omega) = \begin{cases} 1, & \omega \in A, \\ 0, & \omega \notin A. \end{cases}$$

|             |          |        |
|-------------|----------|--------|
| Value       | 0        | 1      |
| Probability | $P(A^c)$ | $P(A)$ |

Compute its probability-weighted average.

---

---

### Indicator identity

$$\mathbb{E}[I_A] = P(A).$$

For  $I \sim \text{Bern}(p)$ , this gives  $\mathbb{E}[I] = p$ , including  $p = 0$  and  $p = 1$ .

## Mean of a binomial count

Let  $A_i$  be success on trial  $i$ , with  $P(A_i) = p$ , and put  $I_i = I_{A_i}$ .

$$N = I_1 + \cdots + I_n.$$

Use linearity and the indicator identity.

---

---

---

### Two assumptions to distinguish

The mean is  $np$  whenever all  $P(A_i) = p$ .

Mutually independent Bernoulli trials give the distribution  $N \sim \text{Bin}(n, p)$ .

More generally,  $\mathbb{E}[N] = \sum_i P(A_i)$  even with dependence or unequal success probabilities.

## Sampling without replacement

### A count with dependent indicators

A bag has three red balls and two blue balls. Draw two balls uniformly without replacement. Let  $N$  count red balls,  $A_i = \{\text{draw } i \text{ is red}\}$ , and  $I_i = I_{A_i}$ .

- 1 Express  $N$  in terms of  $I_1, I_2$  and find its mean.
  - 2 Explain why the draws are dependent.
- 
- 
- 
- 

Each draw has unconditional red probability  $3/5$ . The conditional probability changes.

## Expectation: summary

---

| Object or operation       | Result                                                    |
|---------------------------|-----------------------------------------------------------|
| Discrete mean             | $\mathbb{E}[X] = \sum_x x p_X(x)$                         |
| Finite linear combination | $\mathbb{E}[\sum_i a_i X_i] = \sum_i a_i \mathbb{E}[X_i]$ |
| Indicator                 | $\mathbb{E}[I_A] = P(A)$                                  |
| Count of events           | $\mathbb{E}[\sum_i I_{A_i}] = \sum_i P(A_i)$              |

---

Use finite expectations. Linearity does not require independence.

### Examination boundary

Today's expectation material is excluded from Midterm 1.

# Midterm 1 scope

## Required material through October 9

Events and counting; conditional probability, total probability, and Bayes; independence; random variables, PMFs and CDFs; Bernoulli and binomial laws.

## Excluded

Expectation, variance, and joint distributions.

**Before a calculation** Identify the experiment, the event or random variable, and the assumptions.

Retrieval: in a coin experiment, how do an outcome, the event  $\{X = 2\}$ , and the variable  $X$  differ?

## Roles and committees

A student organization has 12 members. Select three different people.

### **Three distinct offices**

President, vice-president, treasurer.

---

---

---

### **A committee of three**

There are no distinct roles.

---

---

---

The population and selection size are the same. Only the role of ordering changes.

## Four items in seven locations

Four labeled items independently choose among seven labeled locations, uniformly.



Find the probabilities of no collision and of at least one collision.

---

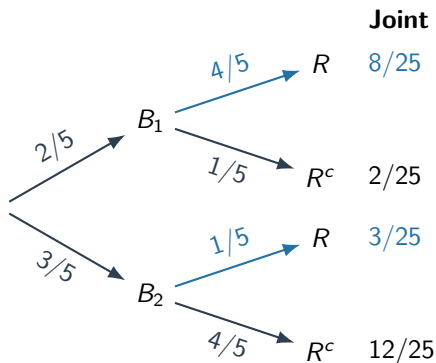
---

---

Independent uniform choices make the  $7^4$  labeled placements equally likely.

## Source probabilities and red outcomes

Select source  $B_1$  with probability  $2/5$ , otherwise  $B_2$ . A result is red with probability  $4/5$  from  $B_1$  and  $1/5$  from  $B_2$ .



**The branch probabilities are conditional**

$$P(R \mid B_1) = \frac{4}{5}, \quad P(R \mid B_2) = \frac{1}{5}.$$

**Explain one path, then add the red paths.**

---

---

---

The sources form a partition. Red can occur through either disjoint source event.

## Posterior probability

The result is red. What is the probability that source  $B_1$  was selected?

| Source | Prior | Red given source | Joint red contribution |
|--------|-------|------------------|------------------------|
| $B_1$  | $2/5$ | $4/5$            | $8/25$                 |
| $B_2$  | $3/5$ | $1/5$            | $3/25$                 |
| Total  |       |                  | $11/25$                |

Restrict to red outcomes and renormalize. State the denominator's meaning.

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

$P(R \mid B_1) = 4/5$  answers a different conditional question.

## Event operations and independence

Suppose  $A, B$  are independent, with  $P(A) = 1/3$  and  $P(B) = 1/4$ .

**Find both, at least one, and neither.**

---

---

---

---

Independent

$$P(A \cap B) = P(A) P(B).$$

Disjoint

$$A \cap B = \emptyset.$$

Disjoint events of positive probability cannot be independent.

If  $P(B) > 0$ , independence is equivalent to  $P(A | B) = P(A)$ . A zero-probability event is independent of every event under the product definition.

## Outcomes, values, and masses

$\Omega = \{1, \dots, 10\}$ , with each outcome having probability  $1/10$ .

**Find  $\{X = 0\}$  and the PMF.**

| Outcomes      | $X(\omega)$                       |
|---------------|-----------------------------------|
| 1, 2          | → <input type="text" value="-2"/> |
| 3, 4, 5, 6, 7 | → <input type="text" value="0"/>  |
| 8, 9, 10      | → <input type="text" value="3"/>  |

---

---

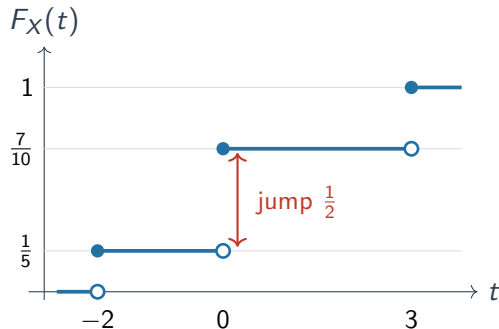
---

### PMF check

The positive-mass values are  $-2, 0, 3$ . Set the PMF to zero elsewhere; its masses must sum to one.

## Masses and CDF jumps

$$F_X(t) = P(X \leq t), \quad t \in \mathbb{R}.$$



|          |       |       |        |
|----------|-------|-------|--------|
| $x$      | $-2$  | $0$   | $3$    |
| $p_X(x)$ | $1/5$ | $1/2$ | $3/10$ |

Find  $P(X = 0)$ ,  $P(X < 0)$ ,  $P(X \leq 0)$ ,  
and  $P(-2 < X \leq 3)$ .

---

---

---

$$P(a < X \leq b) = F_X(b) - F_X(a), \quad a < b.$$

A PMF mass is a CDF jump. The closed dot gives the CDF value at a jump.

## Binomial probabilities

Eight independent test runs each detect a particular defect with probability  $1/4$ . Let  $N$  count detections. This is a hypothetical model.

$$N \sim \text{Bin}(8, 1/4), \quad N \in \{0, 1, \dots, 8\}.$$

**Exactly two detections**

---

---

---

**At least one detection**

---

---

---

Check fixed  $n$ , two outcomes per trial, common  $p$ , and mutual independence.

Pairwise independence alone does not justify the probability of a full pattern. Exact expressions suffice.

## Review: final checks

---

| Check             | What to identify                                                                 |
|-------------------|----------------------------------------------------------------------------------|
| Model             | Which outcomes are equally likely? Is independence given?                        |
| Event or variable | What precisely is being counted or measured?                                     |
| Counting          | Do order, labels, and replacement matter?                                        |
| Conditioning      | Which event defines the denominator? Is its probability positive?                |
| Distribution      | What is the support? Do the masses sum to one?                                   |
| CDF               | Which endpoints are included? Is the request a mass or a cumulative probability? |

---

Finish with the probability of the requested event. It must lie in  $[0, 1]$ .

# Midterm 1

## Examination meeting

Friday, October 16, 2026

55 minutes of working time within the scheduled class period.

## Scope

Required material through October 9: events, counting, conditioning, independence, random variables, PMFs/CDFs, and Bernoulli/binomial laws.

## Excluded

Expectation, variance, and joint distributions.

## Materials

Instructor-provided formula sheet and a basic non-programmable calculator. No other notes, communication, or electronic resources.

Date follows the current syllabus. Approved accommodations are handled separately.

## Optional: expectation of a function

For finite-valued  $X$  and a real-valued function  $g$ , put  $Y = g(X)$ . Grouping inputs that give the same output gives

$$\mathbb{E}[Y] = \sum_y y \sum_{x:g(x)=y} p_X(x) = \sum_x g(x)p_X(x).$$

**The probabilities stay attached to the original inputs.**

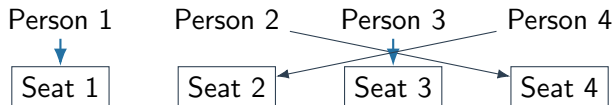
|              |     |     |     |     |
|--------------|-----|-----|-----|-----|
| $x$          | 0   | 1   | 2   | 3   |
| $p_X(x)$     | 1/8 | 3/8 | 3/8 | 1/8 |
| $g(x) = x^2$ | 0   | 1   | 4   | 9   |

---

For countably supported  $X$ , assume  $\sum_x |g(x)|p_X(x) < \infty$ . This extension is excluded from Midterm 1 and will be revisited before variance.

## Optional: random matching

There are  $n \geq 1$  labeled seats and  $n$  people, each with an assigned seat. Choose one of the  $n!$  seating permutations uniformly. Let  $M$  count correct matches.

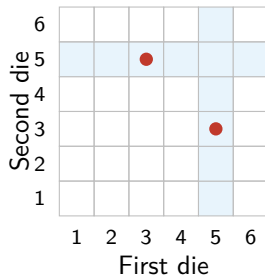


Let  $I_i$  indicate that person  $i$  gets seat  $i$ . Find  $\mathbb{E}[M]$ .

The diagram shows one seating of four people. The mean calculation covers all permutations.

## Optional: changed conditioning

Roll two independent fair dice. Let  $S$  be their sum and  $B$  the event that at least one die shows 5.



### Compare three observations

- ① At least one die shows 5.
- ② The first die shows 5.
- ③ Exactly one die shows 5.

What is the probability of  $S = 8$  under each observation?

---

---

---

Count favorable outcomes inside the conditioning event.

# References for this week

## DeGroot and Schervish

*Probability and Statistics*, fourth edition.

- §4.1, pp. 207–209: discrete expectation and Bernoulli example.
- §4.2, pp. 217–221: affine changes, linearity, and sampling means.

## Mor Harchol-Balter

*Introduction to Probability for Computing*, supplied 2024 version.

- §4.1, pp. 58–59: opening expectation definition and examples.
- §4.2, pp. 63–64: linearity and the binomial mean.
- Example 4.12, pp. 64–65: optional random matching.

Selections only. The lecture gives a finite-outcome proof in place of the books' joint-law or continuous arguments. Review references are the current Weeks 1–2 slides.

Additional source: Sheng Jiang, STAT 131 Notes, July 18, 2024, selected statements on pp. 48–51. Examples adapt Antonio de Leon's earlier STAT 131 sources.